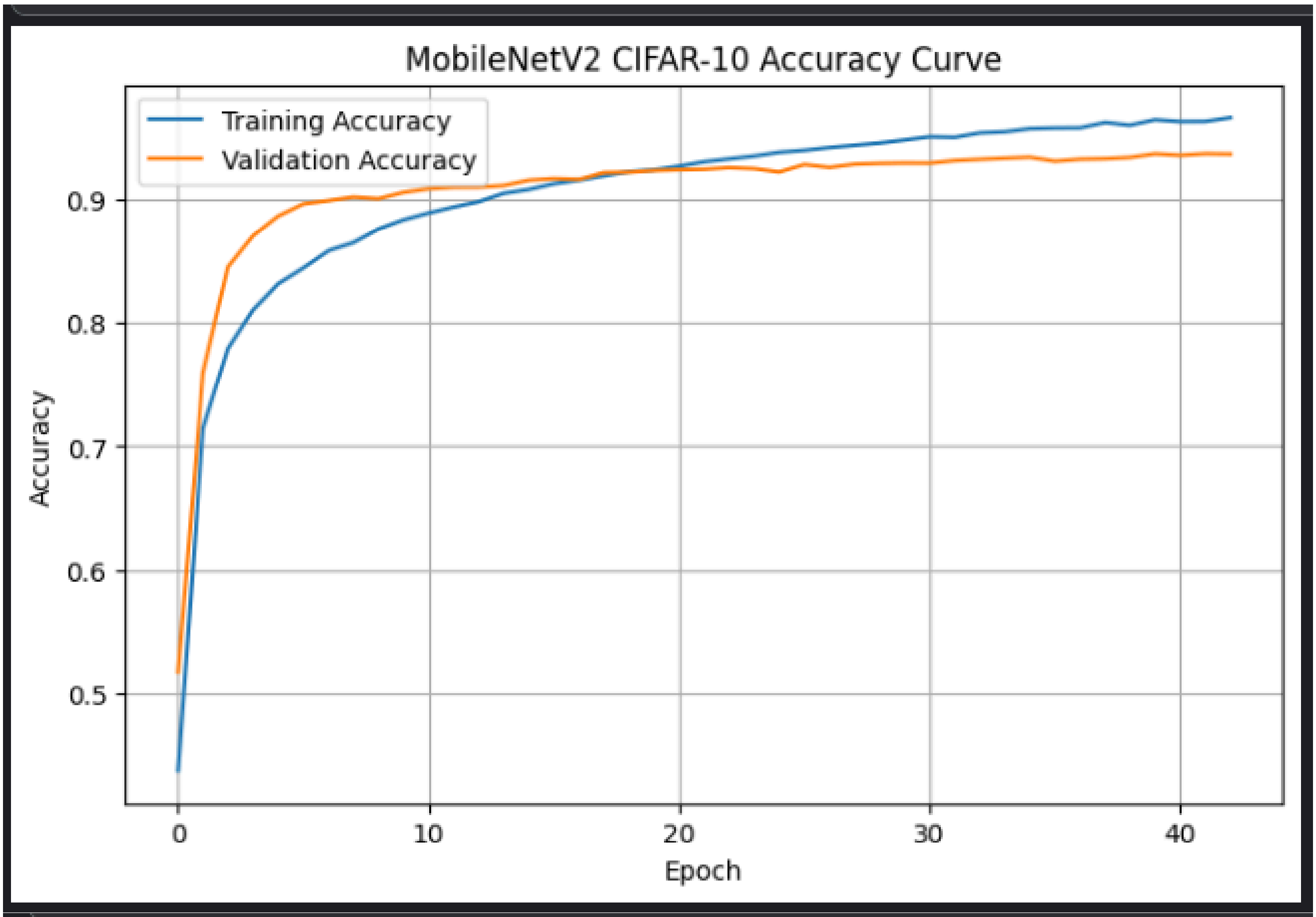
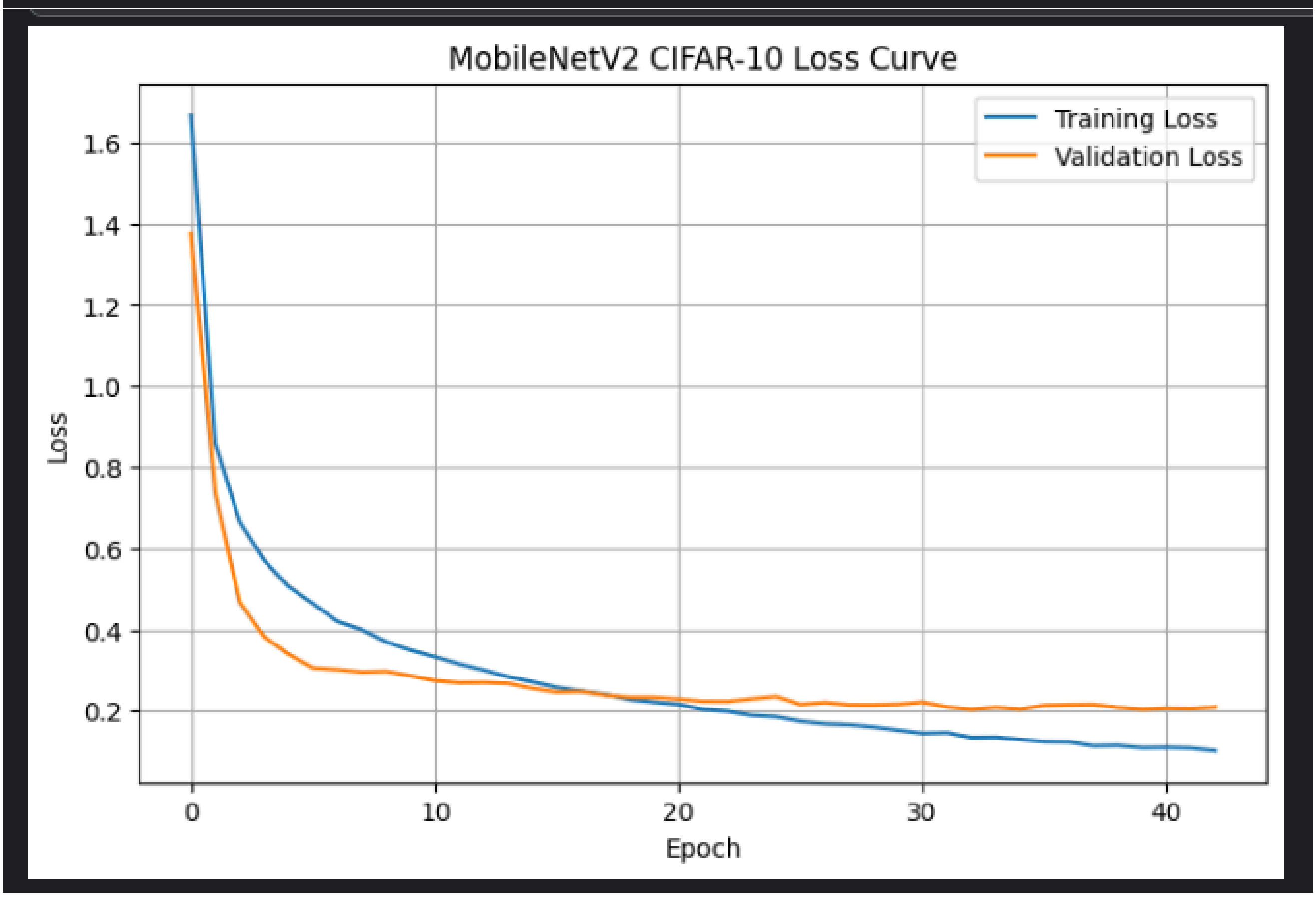
MOBILENETV2 CIFAR-10 DATASET

Visualizing the Learning Curves





MOBILENETV2 CIFAR-10 COMPLETE PROJECT SUMMARY

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            MobileNetV2 CIFAR-10 COMPLETE PROJECT SUMMARY
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DATASET INFORMATION
--------------------------------------------------------------------------------
Dataset Name             : CIFAR-10
Number of Classes        : 10
Test Samples             : 10000

MODEL PERFORMANCE
--------------------------------------------------------------------------------
Training Accuracy        : 96.53%
Validation Accuracy      : 93.62%
Overall Accuracy         : 93.57%
Test Accuracy            : 93.57%
Test Loss                : 0.2089

PERFORMANCE METRICS
--------------------------------------------------------------------------------
Precision                : 93.54%
Recall                   : 93.57%
F1 Score                 : 93.54%

MODEL INFORMATION
--------------------------------------------------------------------------------
Total Parameters         : 2,588,490
Model Size               : 30.05 MB

INFERENCE PERFORMANCE
--------------------------------------------------------------------------------
Total Inference Time     : 0.472944 sec
Average Time Per Image   : 0.472944 sec
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PROJECT PERFORMANCE SUMMARY

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                  PROJECT PERFORMANCE SUMMARY
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DATASET INFORMATION
------------------------------------------------------------------------------------------
Dataset Name               : CIFAR-10
Number of Classes          : 10
Training Samples           : 40000
Validation Samples         : 10000
Test Samples               : 10000

MODEL PERFORMANCE
------------------------------------------------------------------------------------------
Training Accuracy          : 96.53%
Validation Accuracy        : 93.62%
Test Accuracy              : 93.57%
Test Loss                  : 0.2089

PERFORMANCE METRICS
------------------------------------------------------------------------------------------
Precision                  : 93.54%
Recall                     : 93.57%
F1 Score                   : 93.54%

MODEL INFORMATION
------------------------------------------------------------------------------------------
Model Name                 : MobileNetV2
Total Parameters           : 2,588,490
Model Size                 : 30.05 MB

INFERENCE PERFORMANCE
------------------------------------------------------------------------------------------
Total Inference Time       : 0.472944 sec
Average Time Per Image     : 0.472944 sec

EXTERNAL IMAGE TESTING
------------------------------------------------------------------------------------------
External Images Tested     : 10
Correct Predictions        : 9
Incorrect Predictions      : 1
External Accuracy          : 90.00%
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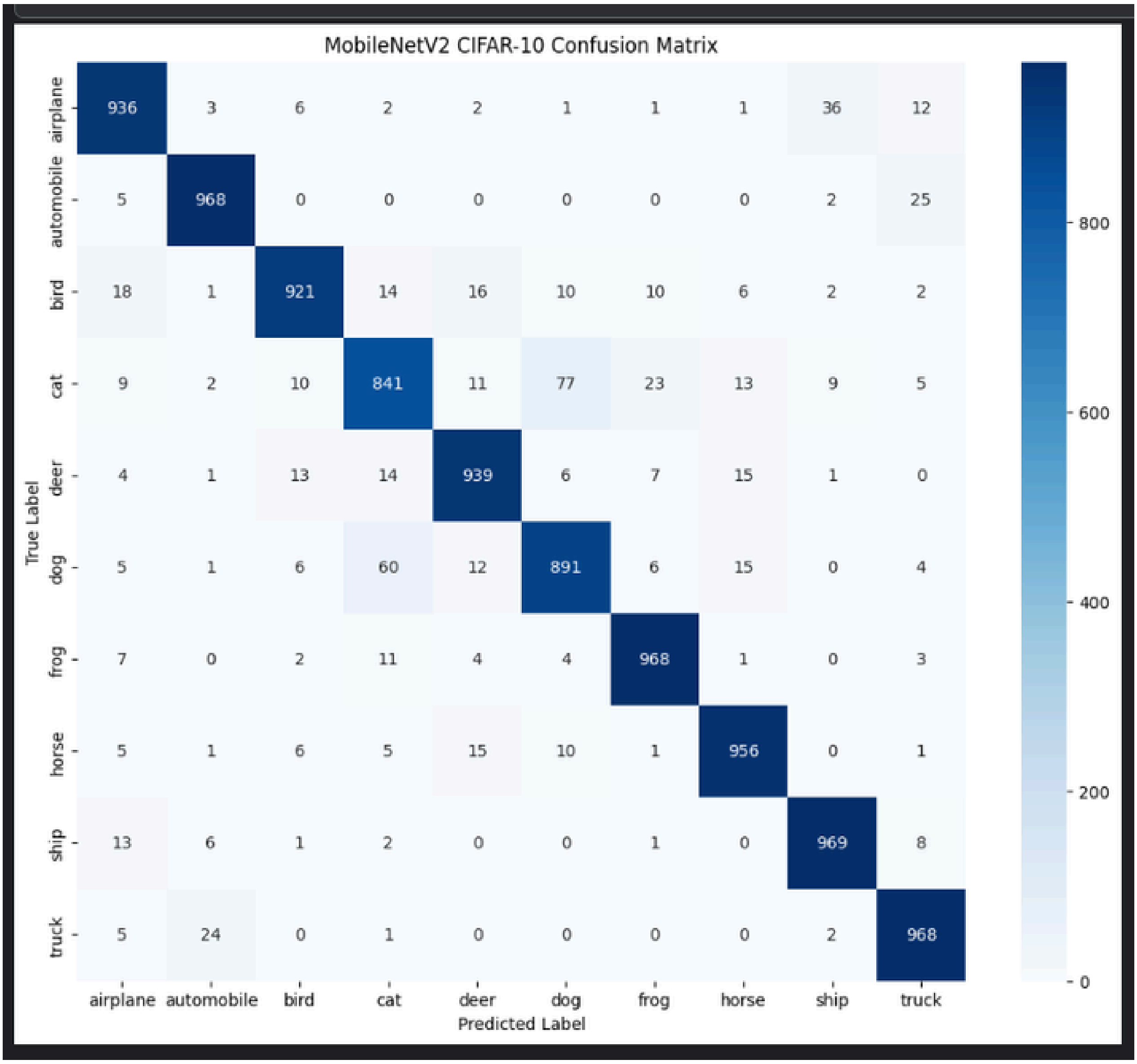
CONFUSION MATRIX

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                 CONFUSION MATRIX SUMMARY
=====================================================================================================
Dataset Name               : CIFAR-10
Number of Classes          : 10
Test Samples               : 10000

Overall Accuracy           : 93.57%
Test Loss                  : 0.2089

Correct Predictions        : 9357
Incorrect Predictions      : 643
Confusion Matrix Accuracy  : 93.57%
Confusion Matrix Status    : Excellent Classification Performance

OBSERVATION
-----------------------------------------------------------------------------------------------------
The confusion matrix shows strong
classification performance across
most CIFAR-10 classes with very
few misclassifications.
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CONFUSION MATRIX SUMMARY

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CONFUSION MATRIX SUMMARY
--------------------------------------------------------------------------------
Dataset                 : CIFAR-10
Number of Classes       : 10
Test Samples            : 10000
Overall Accuracy        : 93.57%
Test Loss               : 0.2089

Correct Predictions        : 9357
Incorrect Predictions      : 643
Total Test Images          : 10000
Confusion Matrix Accuracy  : 93.57%
Confusion Matrix Status    : Excellent Classification Performance
Confusion Matrix Generated : Yes

GENERATED OUTPUTS
--------------------------------------------------------------------------------
✓ Accuracy Curve
✓ Loss Curve
✓ Classification Report
✓ Performance Metrics
✓ Confusion Matrix
✓ Project Summary

SAVED FILES
--------------------------------------------------------------------------------
✓ best_mobilenetv2.keras
✓ mobilenetv2_cifar10.keras
✓ accuracy_curve.png
✓ loss_curve.png

PROJECT STATUS
--------------------------------------------------------------------------------
✓ Training Completed
✓ Evaluation Completed
✓ Testing Completed
✓ Model Saved
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EXTERNAL IMAGE TESTING SUMMARY

EXTERNAL IMAGE TESTING SUMMARY

Actual Class	Predicted Class Index	Predicted Label	Confidence (%)	Prediction Status	Result
Ship	8	Ship	99.83	Very High Confidence	Correct
Airplane	0	Airplane	99.34	Very High Confidence	Correct
Automobile	1	Automobile	54.67	Moderate Confidence	Correct
Bird	2	Bird	99.92	Very High Confidence	Correct
Cat	3	Cat	98.72	Very High Confidence	Correct
Deer	4	Deer	31.71	Low Confidence	Correct
Dog	5	Dog	97.45	Very High Confidence	Correct
Frog	3	Cat	92.11	Very High Confidence	Incorrect
Horse	7	Horse	86.55	High Confidence	Correct
Truck	9	Truck	95.11	Very High Confidence	Correct

SUMMARY

Total External Images Tested : 10
Correct Predictions : 9
Incorrect Predictions : 1
External Testing Accuracy : 90.00%

CONFIDENCE ANALYSIS

Average Confidence Score : 85.54%
Highest Confidence Score : 99.92%
Lowest Confidence Score : 31.71%

MISCLASSIFICATION DETAILS

Actual Class : Frog
Predicted Class : Cat
Confidence Score : 92.11%

EXTERNAL TESTING STATUS

✓ 10 External Images Tested
✓ 9 Images Correctly Classified
✓ 1 Image Misclassified
✓ Confidence Analysis Completed
✓ External Testing Completed Successfully

FINAL CONCLUSION

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                         FINAL CONCLUSION
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1. MobileNetV2 was successfully implemented on the CIFAR-10 dataset.

2. The model achieved excellent classification performance
   with a Test Accuracy of 93.57%.

3. Precision, Recall and F1-Score values were above 93%,
   demonstrating balanced and reliable predictions.

4. The model contains approximately 2.59 million parameters,
   making it lightweight and efficient.

5. The model size is compact and suitable for deployment
   on resource-constrained edge devices.

6. External image testing was performed using 10 real-world images.

7. The model correctly classified 9 out of 10 images,
   achieving an External Testing Accuracy of 90%.

8. The Frog image was misclassified as Cat,
   while all remaining classes were predicted correctly.

9. Confusion matrix analysis indicated excellent
   class-wise classification performance.

10. MobileNetV2 demonstrated significantly better
    performance compared to the basic CNN model.

11. The trained model is suitable for:

    • TensorFlow Lite Conversion
    • Quantization
    • ESP32 Deployment
    • TinyML Applications
    • Edge AI Systems

FINAL PROJECT STATUS

✓ Data Preprocessing Completed
✓ Model Training Completed
✓ Model Evaluation Completed
✓ Performance Metrics Generated
✓ Confusion Matrix Generated
✓ External Image Testing Completed
✓ Model Saved Successfully
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